

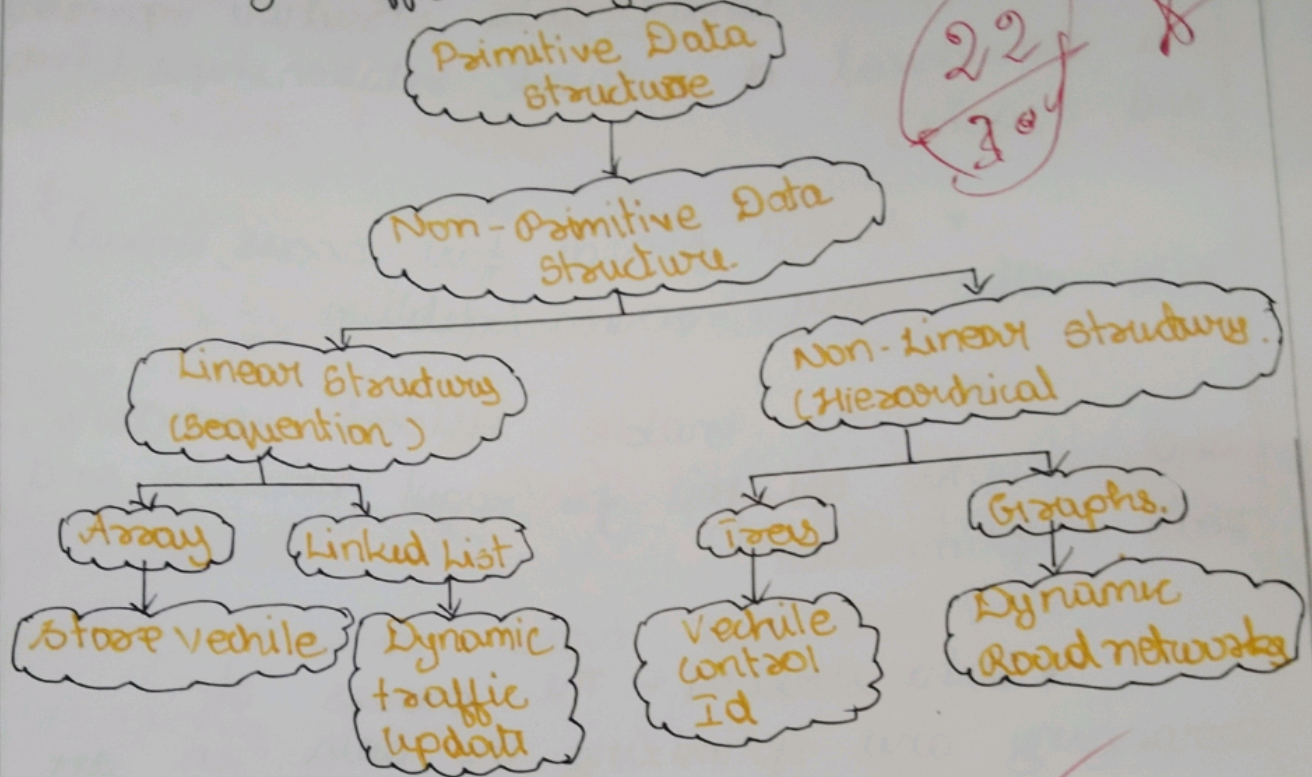
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CSA0303 - Data Structure

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CO1 - AT1 - Concept Maps. test

1. Smart City Traffic Management - Concept Map.



Data structures:

* Data structures are of ~~two~~ types
~~they are~~ Primitive Data structure and
Non-Primitive Data structure.

* The Primitive Data structure stores
the basic values.

* The Non-Primitive Data structure
organize the complex data

* Linear data structure store the sequentially. [Arrays and linked devices.

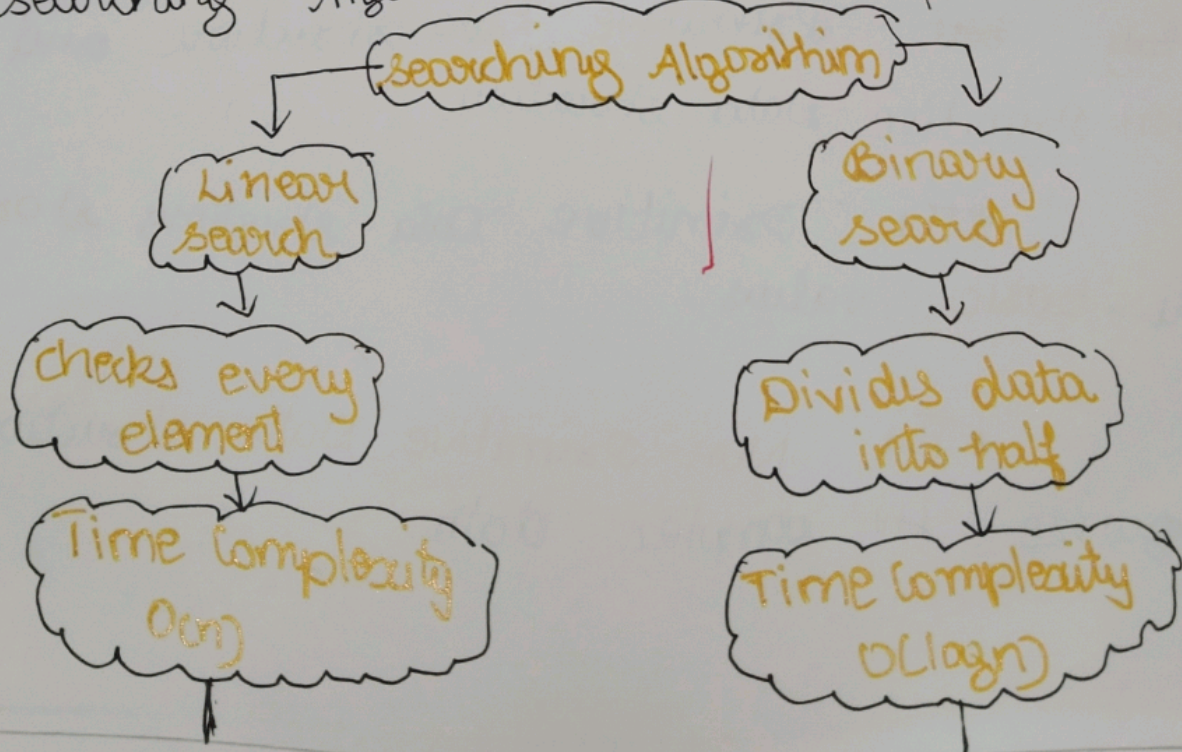
* Non-Linear data structure represent the hierarchical of network relationships (Trees and Graphs).

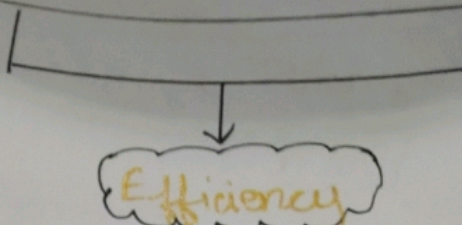
* Arrays provide fast access, linked lists allow easy insertion / deletion.

* Trees enable efficient searching, and graphs are best for social networks and route optimization.

• Data structure - The process of arranging and optimizing the data in a primitive and non-primitive order.

2. Searching Algorithms - Concept Map.





Efficiency

- The Binary search is faster for sorted data.

- * Linear search checks elements one by one.

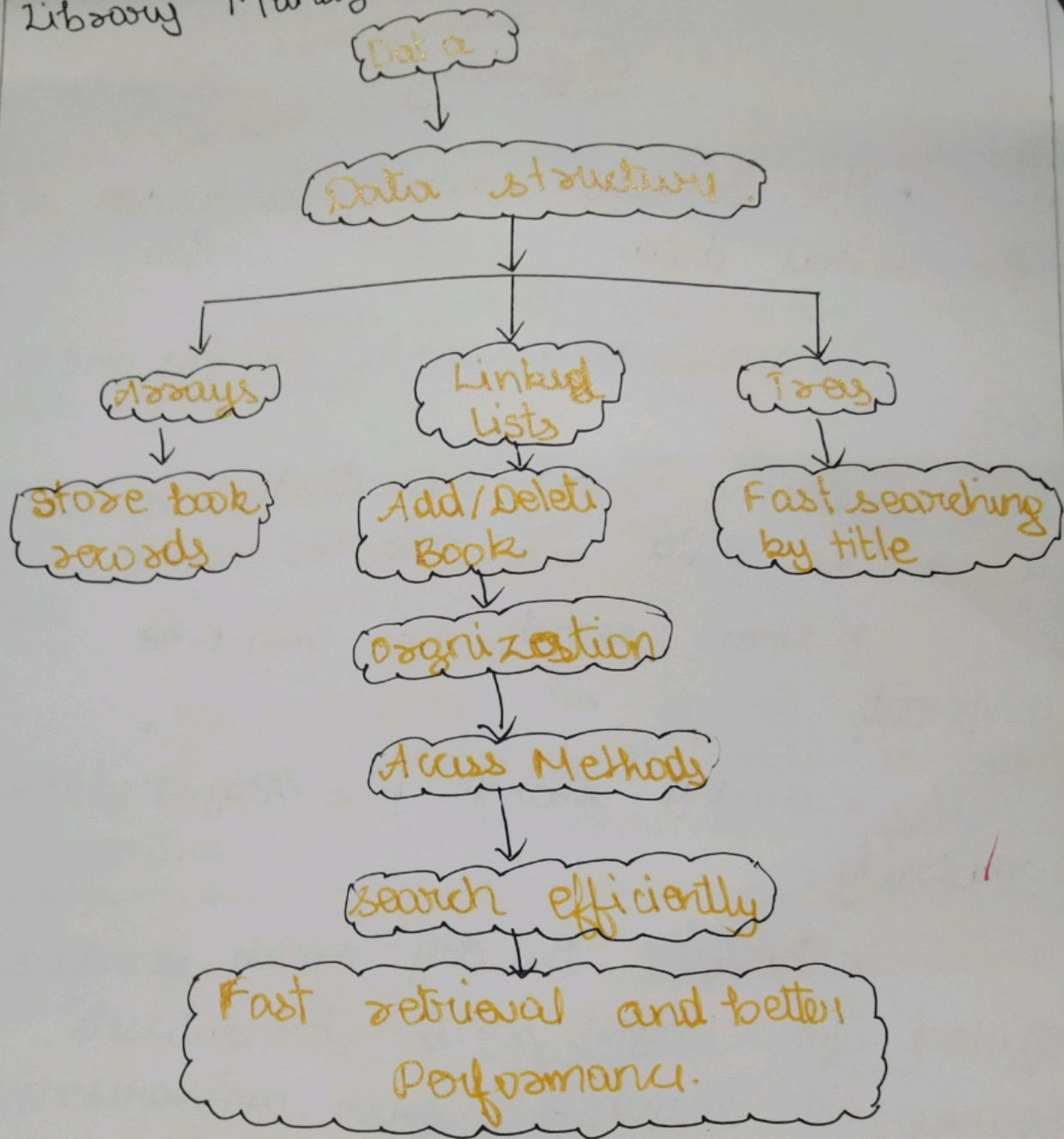
- * Binary search repeatedly divides a sorted data set into two halves.

- * Linear search has $O(n)$ time complexity.

- * Binary search has $O(\log n)$ time complexity.

- * Therefore Binary search is more efficient for large sorted ~~list~~ datasets, because it requires fewer comparisons.

3. Library Management System - Concept Map.



* Data is organized using suitable data structure.

* Arrays store book records efficiently

and * Linked lists support dynamic insertion and deletion.

* Trees enables quick searching of books.

* Proper organization reduces search time, improves retrieval speed, and enhances overall system performance.
